AG-RACE: Antigravity Training & Hover Physics

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March 13, 2026

1 Introduction

This document outlines the mathematical foundation for the Antigravity Racing (AG-RACE) environment, specifically focusing on the newly implemented Hovercraft Physics model and the CMA-ES (Covariance Matrix Adaptation Evolution Strategy) training pipeline.

2 Antigravity Hover Physics

The vehicle movement is no longer kinematic. It is governed by separate longitudinal and lateral velocity components, allowing for realistic inertia and drifting.

2.1 Forward Dynamics

The longitudinal velocity v_f is updated via:

$$v_{f,t+1} = (v_{f,t} + \alpha_{max} \cdot \tau \cdot \Delta t) \cdot \gamma_{long} \cdot \delta_{hover}$$

Where:

- α_{max}: Maximum acceleration force.
- $\tau \in [0, 1]$: Throttle input.
- γ_{long}: Longitudinal friction (damping).
- δ_{hover}: Hover stabilization factor.

2.2 Lateral Dynamics (Drifting)

Lateral velocity v_l is governed by:

$$v_{l,t+1} = (v_{l,t} - \Delta\theta \cdot v_f \cdot \kappa_{inertia}) \cdot \gamma_{lat}$$

Where:

- $\Delta\theta$: Change in rotation (heading) per frame.
- $\kappa_{inertia}$: Centrifugal inertia coefficient.
- γ_{lat}: Lateral friction, modified by *drift* and *stabilize* inputs.

3 Training via CMA-ES

The `NeuralAgent` is trained using the CMA-ES algorithm, which is highly efficient for non-differentiable black-box optimization in continuous parameter spaces.

3.1 Fitness Function

The objective is to maximize the mathematical fitness F:

$$F = (\text{Progress} \times 1000) + (\text{Reward} \times 10) - (\text{Collisions} \times 50) - (\text{OffTrack} \times 1)$$

This ensures the agent learns to:

1. Progress as far as possible on the track.
2. Avoid walls (permadeath penalty).
3. Maintain stability on the asphalt.

3.2 Neuroevolution

The policy network π_ω maps the 59-dimensional observation space to 5 control outputs. CMA-ES optimizes the weights ω by adapting a multivariate normal distribution based on the performance of the population in each generation.